older stars are poorer in hydrogen than younger ones, and that this fact points to a gradual consumption of hydrogen.

Now why is it that we have been unable to produce similar quantities of energy in the laboratory before the discovery of the fission of uranium, assuming for the purposes of this question that we had at our disposal a very powerful source of neutrons, consisting of about 100 grammes of radium mixed with beryllium? Compared with the sources normally available in the laboratory, this would be a very plentiful one, indeed. If we were to use this source of radiation to irradiate ordinary table salt (sodium chloride) for a whole day, an estimated 20,000 million chlorine atoms would be changed into radioactive sulphur atoms. This is a very high number, and the sulphur produced would be actually very strongly radioactive. But unfortunately, this quantity of sulphur is extremely small—just about one thousandth of a millioneth of a milligram. The energy obtained from it is correspondingly small in quantity—just six-millionths of a kilocalorie.

However, the largest cyclotron in operation at the present time can increase the neutron intensity to a point approximately a thousand times what used to be regarded as the upper limit. This intensity equals roughly that of a neutron source consisting of 100 kilograms of radium mixed with beryllium. In this case, both the quantity of the substance and the energy have been multiplied by approximately 1,000, and yet they remain extraordinarily small. At any rate, the energy yield is still just a minute fraction of the energy input required by the cyclotron in order to achieve this result. Therefore, a practical exploitation of nuclear energy will be possible only when the transmutation reactions take place spontaneously—and when such a reaction induces a second one and that one induces still another, and so forth, so that finally, as in chemical reactions, most of the nuclei have become transmuted in a *chain reaction.*

II. URANIUM FISSION AND CHAIN REACTION

In 1938, Hahn and Strassman discovered the process of fission in the uranium nucleus, which process has already been described on pages 134 and 135. A uranium nucleus is hit by a neutron, splits into two approximately equal parts, and in most instances, several neutrons are hurled forth. This reaction constitutes the foundation of modern nucleonics, since it permits the occurrence of the chain reaction, mentioned at the end of the preceding paragraph.

Now a few more details of this process of splitting, or fission, of nuclei. It releases a great deal of energy, in other words, the two fragments of the nucleus are hurled apart with enormous velocity. This energy—about 150